METRIC

DOD-P-51463(EA)
12 February 1980

MILITARY SPECIFICATION

PHOSPHORUS, RED, OILED, TECHNICAL (METRIC)

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one technical grade of oiled red phosphorus.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71 - Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
VV-L-820 - Lubricating Oil, General Purpose (Light)

MILITARY

MIL-P-211 - Phosphorus, Red, Technical (Metric)
MIL-B-82647 - Bag, Conductive Plastic, Heat-Sealable, Flexible

FSC 1365

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads for 40" by 48" Pallets

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171 to 179 - Department of Transportation Hazardous Materials Regulations.

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders for the above publication should cite "49 CFR 171 to 179.")

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

E11 - Wire-Cloth Sieves for Testing Purposes

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Composition. Oiled red phosphorus shall consist of red phosphorus conforming to class 3 of MIL-P-211 which has been uniformly coated with lubricating oil conforming to VV-L-820. No other material shall be used in the manufacture of the oiled red phosphorus. The contractor shall certify that this requirement has been met.

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3.2 Appearance. Oiled red phosphorus shall be free from foreign matter and visible impurities when tested as specified in 4.2.4.1.

3.3 Chemical and physical characteristics. Oiled red phosphorus shall conform to the chemical and physical characteristics of table I when tested as specified therein.

TABLE I. Chemical and physical characteristics

Characteristic	Percent by weight	Test paragraph
Oil content	1.22 to 1.28	4.2.4.2
Volatile matter, maximum	0.25	4.2.4.3
Particle size: retained on a No. 60 (250 micrometer) sieve, maximum	0.5	4.2.4.4

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the oiled red phosphorus produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. No more than one lot of red phosphorus and one lot of lubricating oil shall be used to produce the lot of oiled red phosphorus. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

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4.2.2.2 For test. Sampling shall be conducted in accordance with table II. A representative specimen of approximately 50 grams (g) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE II. Sampling for test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled shipping container ready for shipment. Sample shipping containers shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Bag liner missing or not as specified
(f) Marking incorrect, missing, or illegible
(g) Evidence of oiled red phosphorus on bag liner exterior
(h) Unitization not as specified (when required)

4.2.3.2 For test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.4 Tests. Reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Appearance. Visually examine the specimen for evidence of foreign matter and impurities.

4.2.4.2 Oil content. Weigh to the nearest milligram approximately 5 g of the specimen into a 100-milliliter (ml) beaker. Add 50 ml of methylene chloride and stir for 30 minutes with a mechanical stirrer. Filter through a double thickness of 12.5-centimeter Whatman No. 42 or equal filter paper

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into a clean, tared, glass evaporating dish. Wash with 50 ml of methylene chloride. Place the evaporating dish containing the methylene chloride and oil on a steam bath and evaporate to approximately 3 ml of solution. Remove the remaining methylene chloride in an air stream. Dry off traces of outside moisture, cool, and weigh. Calculate the percent by weight oil content as follows:

$$\text{Percent oil} = \frac{100\ (A - B)}{W}$$

where: A = Weight of dish and residue in grams,
B = Weight of dish in grams, and
W = Weight of specimen in grams.

4.2.4.3 Volatile matter. Heat a flat-bottomed glass or aluminum dish with a loosely fitting cover in a hot water bath for at least 30 minutes. Cool the dish and cover in a desiccator for 30 minutes and weigh them together. Immediately transfer 5 g of specimen to the dish, cover it, and accurately weigh. Place the dish, with cover removed, in a vacuum desiccator over freshly activated silica gel and maintain a substantial vacuum for 24 hours. Release the vacuum slowly and pass the incoming air through a suitable drying vessel filled with activated silica gel. Replace the dish cover and weigh. Calculate the percent by weight volatile matter as follows:

$$\text{Percent volatile matter} = \frac{100\ (A - B)}{(A - C)}$$

where: A = Weight of dish, cover, and specimen before evaporation, in grams,
B = Weight of dish, cover, and specimen after evaporation in grams, and
C = Weight of empty dish and cover in grams.

4.2.4.4 Particle size. Transfer 10 g of specimen onto a tared No. 60 sieve conforming to ASTM E11. Tap the sieve over a sheet of white paper until no significant amount of material passes through the sieve. Add an additional 10 g of specimen and repeat the sieving procedure. Brush any remaining residue on the sieve with a 25-millimeter varnish brush until no material passes through the sieve. Weigh the sieve and material retained on it. Calculate the percent by weight material retained on the sieve as follows:

$$\text{Percent retained on sieve} = \frac{100\ (A - B)}{W}$$

where: A = Weight of sieve and material retained in grams,
B = Weight of sieve in grams, and
W = Total weight of specimen in grams.

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5. PACKAGING

5.1 Packing, level B. A quantity of 27.20 ± 0.25 kilograms of oiled red phosphorus shall be packed in a removable-head, gasketed-head steel drum which provides moisture resistance and is furnished with an electrically conductive, close-fitting bag liner. The drum shall conform to Department of Transportation (DOT) Specification 6A or 6B. The bag shall conform to MIL-B-82647 and shall be formed from film having a minimum thickness of 0.7620 millimeter. The bag shall be closed by tying or knotting. There shall be no evidence of red phosphorus on the exterior of the bag. The drum shall be closed in accordance with DOT regulations.

5.2 Unitization. When specified (see 6.2), shipping containers shall be palletized in accordance with the applicable requirements of MIL-STD-147. Pallets shall conform to type IV of NN-P-71. A horizontal strap shall be applied to each tier and tie-down straps parallel to the 1.016-meter dimension only of the pallet shall be used. When specified (see 6.2), the pallet and any wood storage aids shall be treated with a wood preservative as specified in NN-P-71.

5.3 Marking. Each container shall be marked in accordance with MIL-STD-129 and DOT regulations and shall bear the special prominent marking: "When emptying, clamp the bag to the bare inner surface of the drum and make good electrical connection tc ground."

6. NOTES

6.1 Intended use. Oiled red phosphorus is intended for use as an ingredient in smoke compositions.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) If unitization and wood preservative are required (see 5.2).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE
DOD-P-51463(EA) Phosphorus, Red, Oiled, Technical (Metric)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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